

MUSTAPHA ABDULLAHI

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RESEARCH INTERESTS

Large Language Models, Adaptive Computation, and Efficient Inference

EDUCATION

Aalto University & Finnish Center for Artificial Intelligence (FCAI) **April 2024 - Present**
PhD Artificial Intelligence *Espoo, Finland*

Supervisors: **Prof. Eric Malmi & Prof. Pekka Marttinen**

Queen Mary University of London **Sept. 2022 - Sept. 2023**
MSc Artificial Intelligence [Computer Vision and Robotics] - Grade: 80% (Distinction) *London, U.K.*

Supervisor: **Prof. Bo Zhao**

Ahmadu Bello University **Oct. 2015 - Nov. 2021**
BEng Computer Engineering - CGPA: 4.90/5.00 (First-class) *Zaria, Nigeria*

Supervisor: **Prof. Habeeb Bello-Salau**

Federal Government College, Kaduna **Sept. 2012 - July 2015**
West African Secondary School Certificate (W.A.S.S.C.E) *Kaduna, Nigeria*

MINOR EDUCATION

Mediterranean Machine Learning Summer School (M2L School), Thessaloniki, Greece. **Aug. - Sept. 2023**

The Cornell, Maryland, Max Planck Pre-doctoral Research School (CMMRS), Germany **Aug. 2023**

Reinforcement Learning Summer School (RLSS), Barcelona, Spain **June - July 2023**

INDUSTRY EXPERIENCE

QuantumBlack AI Labs (McKinsey & Company) **Oct. 2023 - Jan. 2024**
Software Engineer Intern *London, U.K.*

- Developed a one-click CLI tool for effortless deployment of **Kedro** projects on **Databricks** Platform.
- Implemented essential client study feedback as integral components within software production pipelines.

Play4Work **Feb. 2021 - March 2022**
Chatbot Developer *Canada(Remote)*

- Collaborated with a team of 5 to design conversation mappings required to build an interactive chatbot.
- Spearheaded a team of 2 to develop a chatbot with over 700 conversation nodes with Google DialogFlow.
- Reduced 60% of the company's workload of candidate's matching and responses processing.

Huawei **Jan. 2020 - Feb. 2020**
Big Data/Cloud Intern *Lagos, Nigeria*

- Set up and managed FTP Server, WordPress (Windows & Linux), Windows Share Point Server, etc. on Huawei Elastic Cloud Server (ECS).
- Performed Relational Database migration using Cloud Data Migration Tool. Utilised Squoop to migrate data between Apache HDFS, Apache HBase, Apache Hive and MySQL database.
- Performed Data Analysis of Smart IoV solution with over 400,000 data points utilizing HiveQL for offline analysis, Apache HBase for real-time query and retrieval and Spark-SQL to migrate the offline analysis service.

RESEARCH EXPERIENCE

Aalto University & FCAI **August 2026 - Present**
Doctoral Researcher (Advised by: **Prof. Eric Malmi & Prof. Pekka Marttinen**) *Espoo, Finland*

- Developing adaptive test-time computation methods for efficient large language models.

AI & Data-Intensive Systems Group (ADIS), Aalto University & FCAI **April 2024 - July 2026**
Doctoral Researcher (Advised by: **Prof. Bo Zhao**) *Espoo, Finland*

- Co-developed **Firebolt-VL** [1], an efficient vision-language model for resource-constrained deployment, and **EARL** [3], a scalable agentic RL training system for large-scale LLM training.

Large-Scale Data & Systems (LSDS), Imperial College London **June 2023 - Aug. 2023**
Research Intern [Master's Thesis] (Advised by: **Prof. Bo Zhao & Prof. Peter Pietzuch**) *London, U.K.*

- Evaluated tradeoffs among three distribution policies of MindSpore Distributed RL.

AWARDS & ACCOLADES

• ACM SOSP 2025 Student Scholarship & Travel Grant Award	2025
• Finnish Doctoral Program Network in Artificial Intelligence (AI-DOC) funded by Finnish Center for Artificial Intelligence (FCAI).	2024-2027
• Finland Fellowship [for first year doctoral studies at Aalto University]	2024
• Black in AI's Emerging Leaders in AI (ELAI)	2023
• AEGEAN Scholarship - Travel Grant to attend M2L School (one of 10 top students)	2023
• Best Graduating Engineering Student Award, NSE (Nigerian Society of Engineers)	2022
• 2023 Rhodes Scholarship Finalist [West Africa]	2022
• Google DeepMind Scholarship	2022-2023
• QuantumBlack (McKinsey Company)/Office of AI Scholarship	2022-2023
• i-Scholar Initiative (iSI) Awardee	2022
• Best ICT Competition Instructor, Huawei ICT Competition Sub-Saharan Africa	2022
• 1st Place, Dept. of Computer Engineering Final Year Undergraduate Project Exhibition	2021
• 3rd Best Engineering Graduate in Nigeria (Awarded by Committee of Deans of Engineering and Technology of Nigerian Universities (CODET))	2021
• NCS (Nigerian Computer Society) Jim Ovia Scholarship Award	2020-2021
• Grand Prize Winner, Huawei ICT Competition Global Final - Cloud Track	2020
• 1st Place, Huawei ICT Competition Regional Final (Southern Africa) - Cloud Track	2020
• 3rd Place, Huawei ICT Competition National Final - Cloud Track	2019
• Nigeria Higher Education Foundation Scholar (Class of 2019)	2019
• 2nd Runner Up, Professor Ayodele Awojobi Design Competition (PAADC) 3.0	2019
• NNPC/Total National Merit Scholarship Award	2018-2021
• China-Nigeria October 1st Scholarship	2018
• Federal Government Scholarship Award [Undergraduate Award]	2017-2021
• Agbami Special Medical and Engineering Professionals Scholarship Award	2017-2021
• Best in Physics, Federal Government College, Kaduna	2015

PUBLICATIONS

 Google Scholar Profile

* denotes equal contribution.

- [1] Quoc-Huy Trinh*, **Mustapha Abdullahi***, Bo Zhao, and Debesh Jha, “Firebolt-VL: Efficient Vision-Language Understanding with Cross-Modality Modulation,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, Jun. 2026, pp. 3654–3663.
- [2] Lukman Jibril Aliyu, Umar Sani Muhammad, Bilqisu Ismail, Nasiru Muhammad, Almustapha A Wakili, Seid Muhie Yimam, Shamsuddeen Hassan Muhammad, and **Mustapha Abdullahi**, “Evaluating Deep Learning Models for African Wildlife Image Classification: From DenseNet to Vision Transformers,” *Proceedings of Machine Learning Research*, vol. 302, Feb. 2026.
- [3] Zheyue Tan, **Mustapha Abdullahi**, Tuo Shi, Huining Yuan, Zelai Xu, Chao Yu, Boxun Li, and Bo Zhao, “EARL: Efficient Agentic Reinforcement Learning Systems for Large Language Models,” *Workshop on Systems for Agentic AI (SAA)*, in conjunction with the *ACM Symposium on Operating Systems Principles (SOSP)*, Oct. 2025.
- [4] Oluwole Fagbohun, Nwaamaka Pearl Iduwe, **Mustapha Abdullahi**, Adeseye Ifaturoti, and OM Nwanna, “Beyond traditional assessment: Exploring the impact of large language models on grading practices,” *Journal of Artificial Intelligence, Machine Learning and Data Science*, vol. 2, no. 1, pp. 1–8, Feb. 2024.
- [5] **Mustapha Abdullahi**, “An Artificial Intelligence of Things (AIOT) based Assistive Smart Glass for the Visually Impaired,” *Final Year Project Poster, Department of Computer Engineering, Ahmadu Bello University, Zaria*, Oct. 2021, Best Project .

TALKS

Kedro Databricks Deploy, QuantumBlack AI Labs, London, U.K.	Dec. 2023
Beyond Building Machine Learning Models  , Data Science Nigeria, Nigeria	June 2021

ACADEMIC SERVICE

Reviewer

- 2025 MLforSystems Workshop|NeurIPS
- 2025 Muslims in ML Workshop|NeurIPS

TEACHING EXPERIENCE

- Research Advisor**, CS-E4645, Research Project on Data Intensive Computing, Aalto University **Spring 2026**
- Guest Lecturer**, CS-E4780, Managing Data in Agentic AI Systems, Aalto University **October 2025**
Instructor: Prof. Bo Zhao
- Teaching Assistant**, CS-E4780, Scalable Systems and Data Management, Aalto University **Autumn 2025**
- Research Advisor**, CS-E4645, Research Project on Data Intensive Computing, Aalto University **Spring 2025**
- Teaching Assistant**, CS-E4780, Scalable Systems and Data Management, Aalto University **Autumn 2024**

PROJECTS

- Evolving AlexNet CNN Maize Plant Disease Classification** **July 2021 to Oct. 2021**
 - Co-designed a modified Binary Particle Swarm Optimisation (BPSO) algorithm to optimise the structure of the fully connected (FC) layers of CNNs.
 - Developed a Tensorflow code implementation of the AlexNet model
 - Contributed to deriving an optimised model architecture that achieved 98.3% accuracy.

- iGlass** **April 2020 - Oct. 2021**
 - A smart assistive device for the visually impaired for reading out text, object detection, scene description, etc. using Raspberry Pi, AWS Rekognition, and Google Cloud AI.
 - Spearheaded the development of iGlass prototype.

- CompengBot** **March 2021 to April 2021**
 - A fast and efficient chatbot to answer queries of students and the general public related to the computer engineering department. Deployed on the departmental website and Telegram page.
 - Easy and improved access to information about department and school.
 - Quick access to details about courses and lecturers contact.

- Transfer Learning for Classification of Cat, Dog and Pandas** **Dec. 2020**
 - Used densenet-161 as feature extractor and achieved an accuracy of 97.3%.

VOLUNTEERING

- Black in AI - Emerging Leaders in AI Grad Prep Program** **Dec. 2024 - Present**
Academic Mentor *Remote*

- iScholars Initiative (iSI)** **Sept. 2023 - Present**
Academic Mentor *Remote*

- Arewa Data Science Academy** **Feb. 2023 - Present**
Lead Mentor, Deep Learning Fellowship (videos ) *Nigeria, Remote*

- 2026 MenaML Winter School** **Sept. 2025**
Application Reviewer

- 2025 Deep Learning Indaba** **Feb. 2025**
Application Reviewer

- 2023 IEEE International Conference on Robotics and Automation (ICRA)** **May 2023**
Volunteer *London, U.K.*

- Kada Hive Innovation and Tech Hub** **May 2022**
Robotics Instructor - Play Tech Kids Programme *Kaduna, Nigeria*

- Data Science Nigeria** **June 2021**
Data Science Tutor *Zaria, Nigeria*

- Assisted participants in setting up data science software and packages.
- Supported the lead tutor in introducing participants to machine learning tools (such as numpy, pandas and scikit-learn).

- Ahmadu Bello University** **Feb. 2018 – April 2021**
Tutor - Control Engineering (COEN303, COEN407, COEN507) *Zaria, Nigeria*

- Computer Engineering, Ahmadu Bello University** **Aug. 2019**
Instructor - Python Bootcamp for Kids *Zaria, Nigeria*

TECHNICAL SKILLS

Programming Languages: Python, $\text{\LaTeX}$; *previously*: Scala, C++, MATLAB, ROS

ML Frameworks: PyTorch, Tensorflow, Keras, Kedro, Apache-Spark, Scikit-Learn, XGBoost

Cloud Services: GCP, AWS, Databricks Platform, Huawei Cloud Platform

Operating System: Linux (Ubuntu, Cent OS 7.*), Windows

Big Data Tools: Hadoop, Apache Spark, Apache Hive, Hbase, MySQL, Huawei MapReduce Service, Gremlin

Microprocessors: Raspberry pi (zero, 3), Arduino, ESP32, ESP8266

Soft Skills: Leadership, Problem Solving, Team Player, Time Management, Interpersonal Skills

PROFESSIONAL MEMBERSHIPS

- Black in AI, member
- Black in Robotics, member
- Nigerian Computer Society (NCS), member
- Data Science Nigeria, member
- Nigerian Society of Engineers (NSE), Graduate Member

HOBBIES

Cooking; watching anime; playing football & table tennis; talks about AI and the future